

Birjung Thapa

Healthcare Data Analyst | Healthcare Analytics | SQL | Python

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🐙 GitHub: github.com/BIRJUNG | 🔗 LinkedIn: linkedin.com/in/birjungthapa | 📁 Portfolio: healthcare-risk-command-center.streamlit.app

PROFESSIONAL PROFILE

Healthcare data analyst and analytics professional with 15+ years of experience translating complex operational data into executive dashboards, forecasting models, KPI reporting, and business recommendations. Built healthcare analytics products for risk adjustment, suspect-member prioritization, claims intelligence, revenue cycle denials, provider performance, PMPM/utilization trends, FWA review, and value-based care scorecards. Hands-on with Python, SQL, Power BI, Tableau, Streamlit, SQLite/DuckDB warehouses, data-quality checks, BI-ready marts, and ML scoring workflows.

EXPERIENCE

Marketing & Commercial Analyst

Arun Valley | Nov 2015 - Jun 2025 | Doha, Qatar

- Designed executive KPI dashboards tracking cost trends, profitability, and operational performance; applied the same reporting discipline to healthcare PMPM, utilization, revenue leakage, and provider scorecard projects.
- Developed forecasting and regression models for project cost estimation and profit/loss analysis, strengthening risk assessment and leadership scenario planning.
- Built structured workforce and performance datasets with validation rules and metric standardization to improve recurring reporting consistency.

Market Research Analyst

Prudential Management and Marketing W.L.L | Nov 2009 - Nov 2015 | Manama, Bahrain

- Led quantitative and qualitative analysis across three international markets, translating market opportunity, client performance, and service-quality data into expansion recommendations.
- Created recurring reports and visualizations for customer satisfaction, operational efficiency, retention, and service-quality trends for cross-functional stakeholders.
- Analyzed client performance metrics to identify risk patterns and recommend targeted interventions, building a foundation for healthcare work-queue and prioritization analytics.

PORTFOLIO PROJECTS

Healthcare Risk Command Center (Live Demo)

- Built a HIPAA-safe Streamlit risk adjustment app across 5,000 synthetic payer members using RAF, HCC-style documentation, utilization, care gaps, PMPM, member months, plan, and provider-group features.
- Trained Logistic Regression and Random Forest models; best model reached 0.792 ROC AUC, 0.526 average precision, 3.49x top-decile lift, and 74.7% critical-tier suspect rate versus 17.9% baseline.

Healthcare Claims Intelligence & Risk Analytics Platform (Live Demo)

- Built a Medicare-style payer analytics pipeline for claims, member months, PMPM, utilization, high-cost members, readmissions, provider outliers, FWA signals, and HCC/RAF-like risk segments.

Claims Denials, Appeals & Revenue Cycle Intelligence (Live Demo)

- Built a revenue cycle analytics platform with DuckDB bronze/silver/gold/mart/ML/DQ schemas for denials, appeals, underpayment, payer scorecards, and work queues.

Provider Network & Value-Based Care Analytics (Live Demo)

- Converted CMS-style provider, hospital quality, ACO, and market data into peer benchmarks, contracting opportunity marts, quality scorecards, and market strategy views.

EDUCATION

Master of Science in Data Science

University of Colorado Boulder | Expected - December 2026

- Relevant focus: data science, statistics, machine learning, SQL, analytics, and applied modeling.

RECRUITER SNAPSHOT

Healthcare Analytics Focus

Risk adjustment, RAF/HCC-style documentation logic, claims, PMPM, denials, appeals, underpayment, provider benchmarking, FWA, and VBC analytics.

Business + Data Bridge

15+ years converting operational, market, cost, and service-quality data into stakeholder decisions and executive reporting.

Portfolio Evidence

Built healthcare apps with risk scoring, review queues, provider action tables, warehouses, SQL packs, dashboards, QA, and deployment.

TECHNICAL SKILLS

Healthcare Analytics SQL Python

Claims Analytics Risk Adjustment

Revenue Cycle Power BI Tableau

Languages: Python, SQL, R, SAS

Healthcare: Risk Adjustment, RAF, HCC-style Documentation, Claims, PMPM, Utilization, Denials

Analytics: EDA, Forecasting, Regression, Classification, Clustering

BI: Power BI, Tableau, Excel, Google Sheets

ML & Stats: Risk Scoring, Logistic Regression, Random Forest, XGBoost

Tools: Power BI, Tableau, Streamlit, Excel, GitHub

Databases: SQLite, DuckDB, SQL Marts

ANALYTICS WORKFLOW

- Ingest synthetic/custom healthcare data and validate structure.
- Model governed tables, marts, and SQL query packs.
- Score risks, outliers, denials, and prioritization queues.
- Publish BI-ready exports and stakeholder dashboards.

APPLIED METHODS

Risk Adjustment RAF / HCC-style Logic

PMPM / Member Months Denial Risk Scoring

Appeal Success Scoring FWA Review Queues

CREDENTIALS

Education: MS Data Science, University of Colorado Boulder (Expected Dec 2026)

Certifications: Google Advanced Data Analytics (2025), Machine Learning with Python (2024), SQL for Data Science (2025)

Leadership: 15+ years analytics and business decision support; Executive stakeholder communication